

Strong Artificial Intelligence

E Information

Mental Entities

Prof. Dr. Dr. h. c. Robert Grebner

Technical University of Applied Sciences, Würzburg, 2026-04-17

Repetition „Time and Space“

- What is used to define the measurement unit of time?
- What is used to define the measurement unit of space?
- Is time/space relative or absolute? Why?
- Even if time and space are mental entities, how can we get an idea of time and space by impressions or preceptions from the physical world?
- What binds time and space inseparable together?
- How would you explain, that space is not existent?
- How would you explain, that time is only a mental phenomenon?

Workshop „Information“

- Where does Information exist?
- What is Information? Define!
- What is an Information Theory?
- What is the Difference between Information and Knowledge?
- Where is the data?

E Information

1. Impressors

Movers

2. Complexity

Simplifier

3. Information Theory

Explanation

4. Information

Knowledge Enabler

5. Knowledge

Action Enabler

6. Knowledge Gap

Motivation

7. Conclusion

Mental Subject



1. Impressors

Movers

1. Impressors

What we know so far

- World (Everything)
 - Physical World
 - Physical Entities (One Thing) - Statics
 - Motion of physical Entities (Change) - Dynamics
 - Mental World
 - Mental Entities: Representations reference other Entities
 - Reality[primary|secondary|...|subjective]: A big set of (interconnected) Mental Entities
 - Time – Mental Entity
 - Measurement Unit: Second
 - Space – Mental Entity
 - Measurement Unit: Meter

1. Impressors

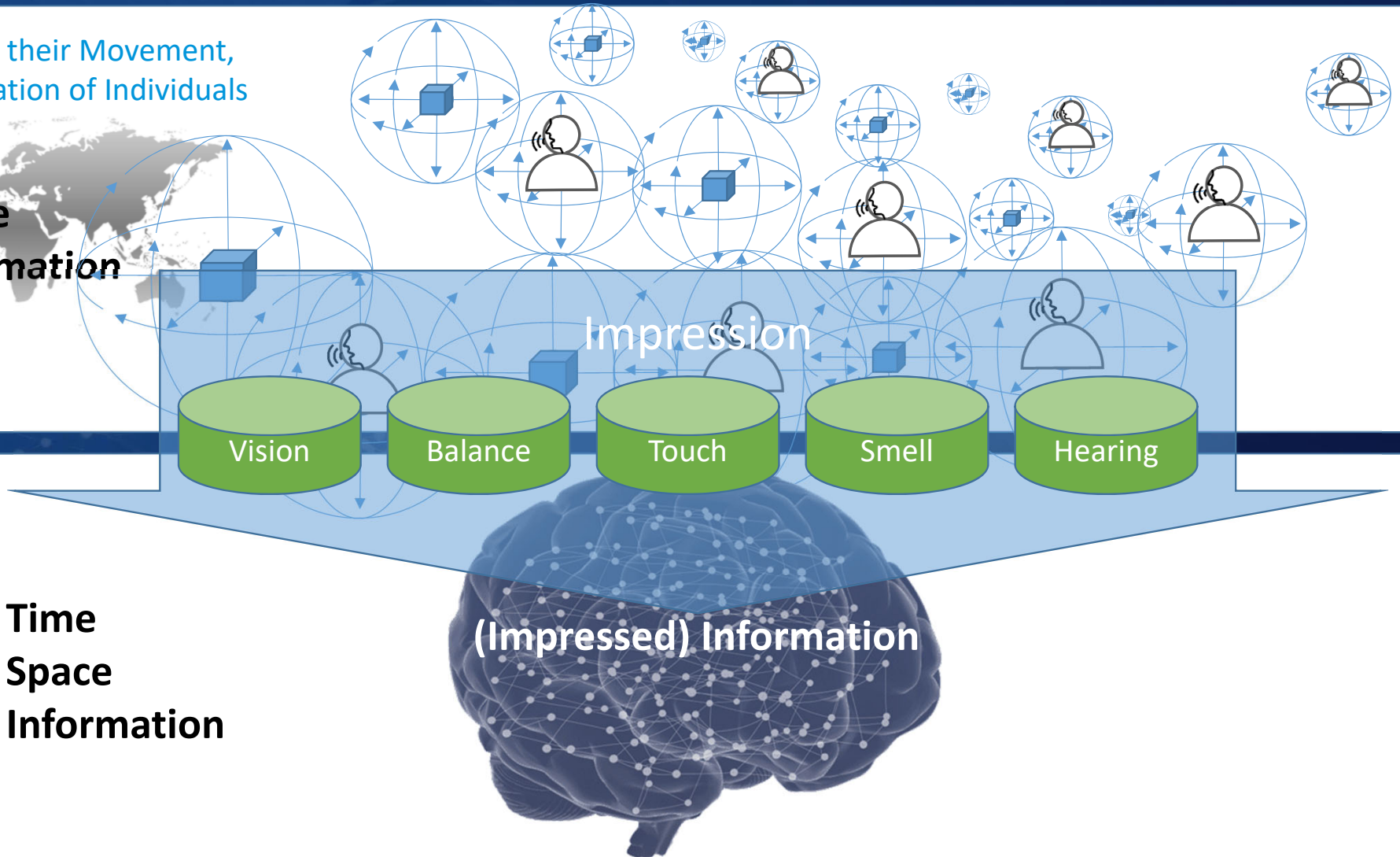
What we can perceive.

Physical World, their Movement,
the Communication of Individuals

No Time
No Space
No Information

Mental World

Artificial Time
Artificial Space
Artificial Information



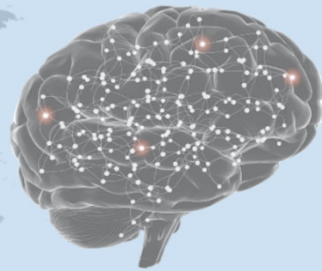
1. Impressors

What we know so far

Physical World

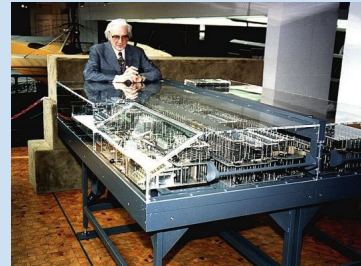
**Systems
Doing
Mental Work
(generating information)**

Homo Sapiens Brain
(about 35.000 BC)



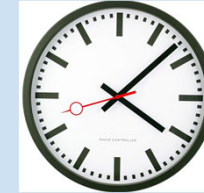
Thinker
(Thought)

Computer
(1937)

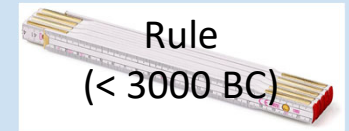


Calculator
(Number)

Clock
(< 3000 BC)



Time Counter
(Second)



Rule
(< 3000 BC)

(> 1960 AD) (> 1993 AD)
Laser Scanner GPS



Distance Counter
(Meter)

Mental World



Information

Knowledge

Thoughts

Data

Information?

Time

Date

Information?

Length, Width, Height,
Position

Area

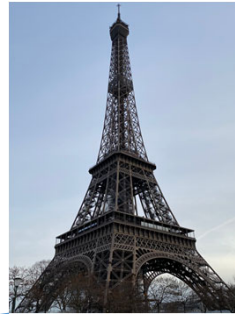
Volume

Information?

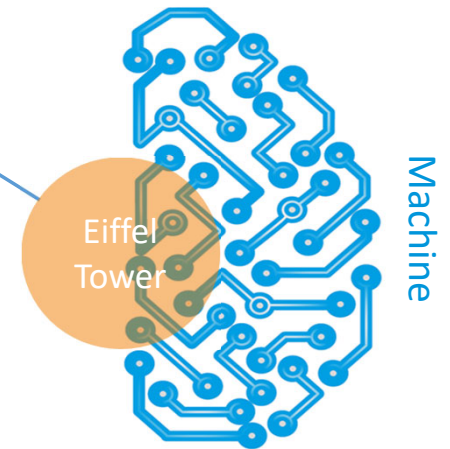
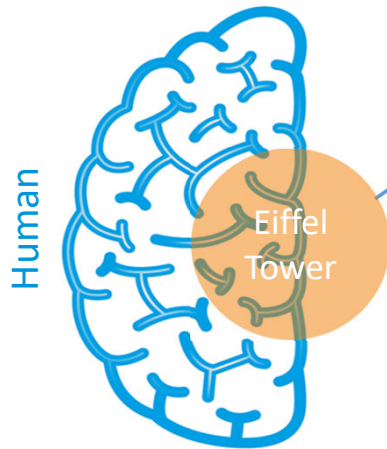
1. Impressors

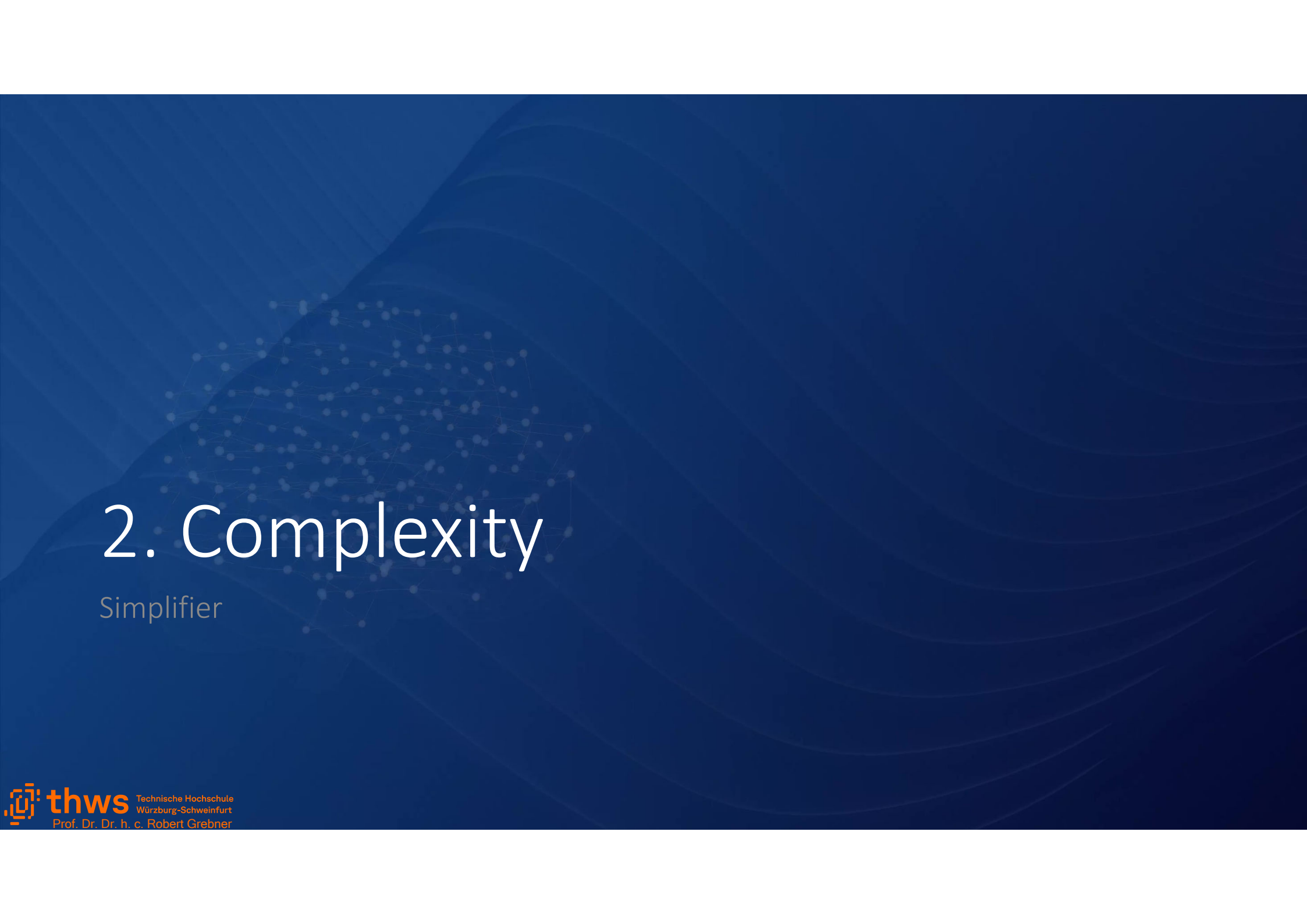
Comparison Human and Machine Perception and Impression

Physical World



Mental World





2. Complexity

Simplifier

Workshop „Complexity“

- Which impressions do we store in our memory and which not?
- What or who decides, about the impressions to be stored?
- What do you think, how impressions are transformed into information?
- How is our mind organising the complexity of all the perceived impressions?
- What constructions is our mind using to simplify or reduce complexity?

2. Complexity

Fundamentals

What is Causing Complexity?

Big Systems with a Lot of Entities **Interacting** and **Moving** in Time and Space.

What is Time Complexity?

When Time is Crucial to Make a System Work or Solve a Problem.

What is Quantity/Space Complexity?

When Space is Crucial to Solve a Problem.

Where is the Complexity?

Inside Brains.

2. Complexity

Methods of Handling

Abstraction?

Using Equivalence Relations
Omitting Details with Focus
on the Essential.

Classification?

Assign Entities to Groups
(narrow/more formal).

Categorisation?

Assign Entities to Groups
(broad/more mental).

Reduction?

Omitting Details without
String Focus on the Essential.



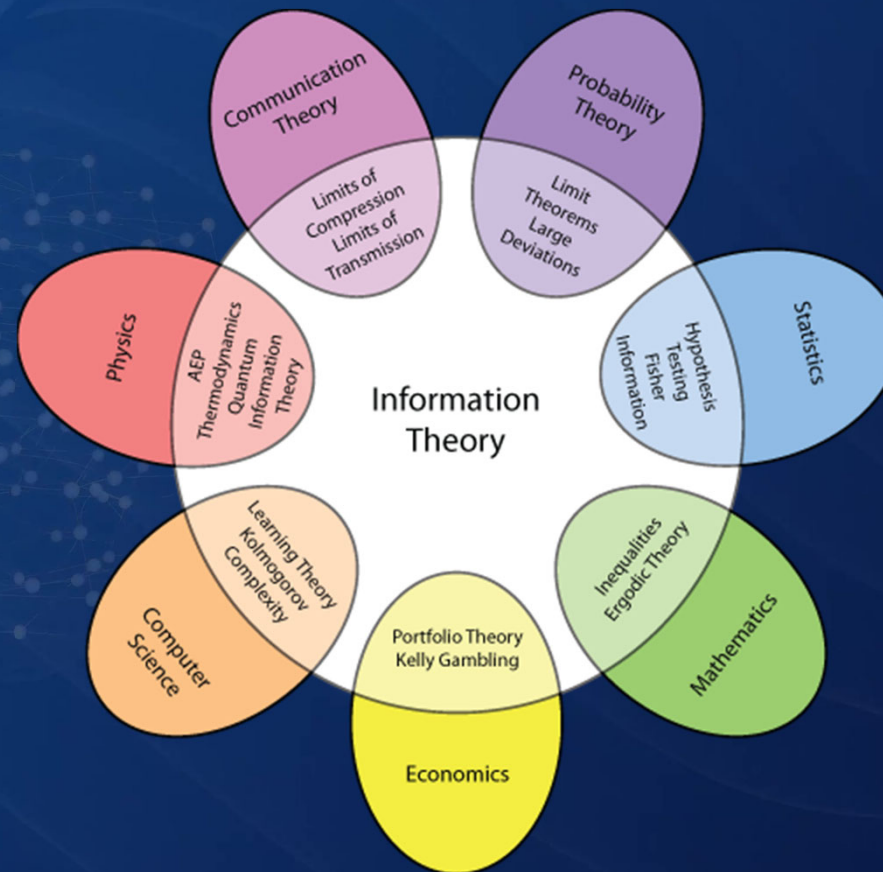
3. Information Theory

Explanation

Workshop „Theory“

- What is the task of a theory and what should a theory for information provide?

3. Information Theory Scientific Disciplines



What is information?

- Information is a mental entity in the mental world
- Information is always part of a brain or thought machine (which itself is part of the physical world – body-mind-dualism)
- Information is always representing other entities
- Information can have tree granularities (atomic, elementary, complex)
- Information processing (logic) and persisting (memory) is task of a brain

What is an entity?

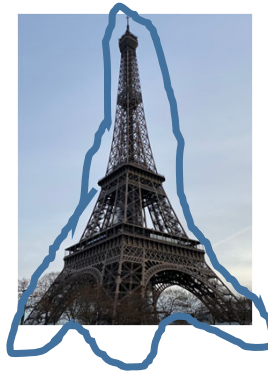
- An entity is everything that is delimitable: No delimitability, no entity
- in the physical world
- in the mental world

3. Information Theory

Entity

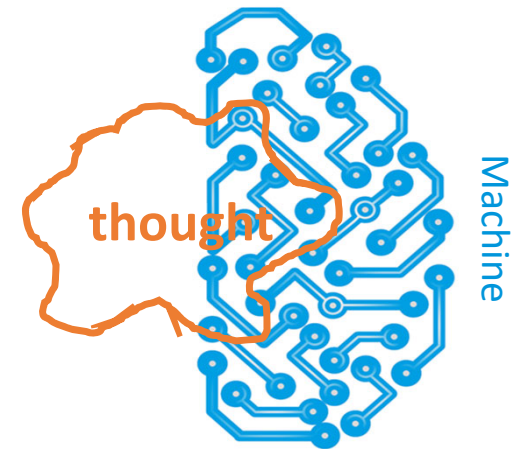
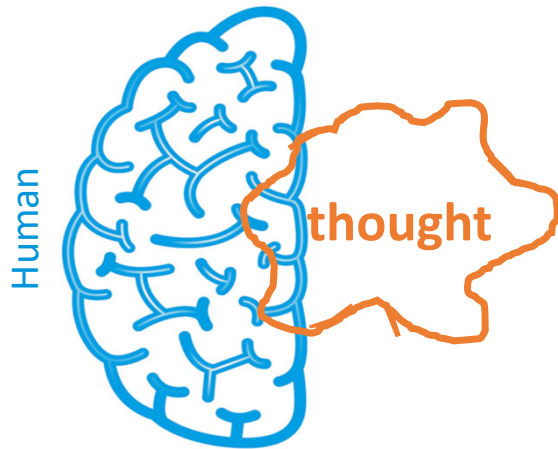
Physical World

delimitable physical something:
physical entity



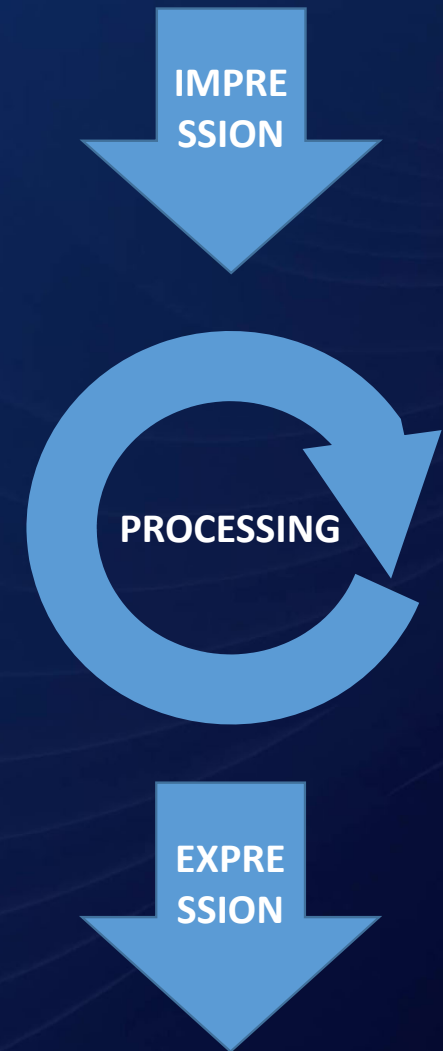
Mental World

delimitable mental something:
mental entity



How is information emerging?

- Information arises through perception (exogenous cause)
 - explicit intentional: learning
 - implicit unintentional: open senses
- Information arises through thinking (endogenous cause)
 - explicit by logic
 - implicit by educated natural neural networks
 - implicit by trained artificial neural networks
- Information can be shared
 - asynchron communication (books, ...)
 - synchron communication (speech, phones, ...)

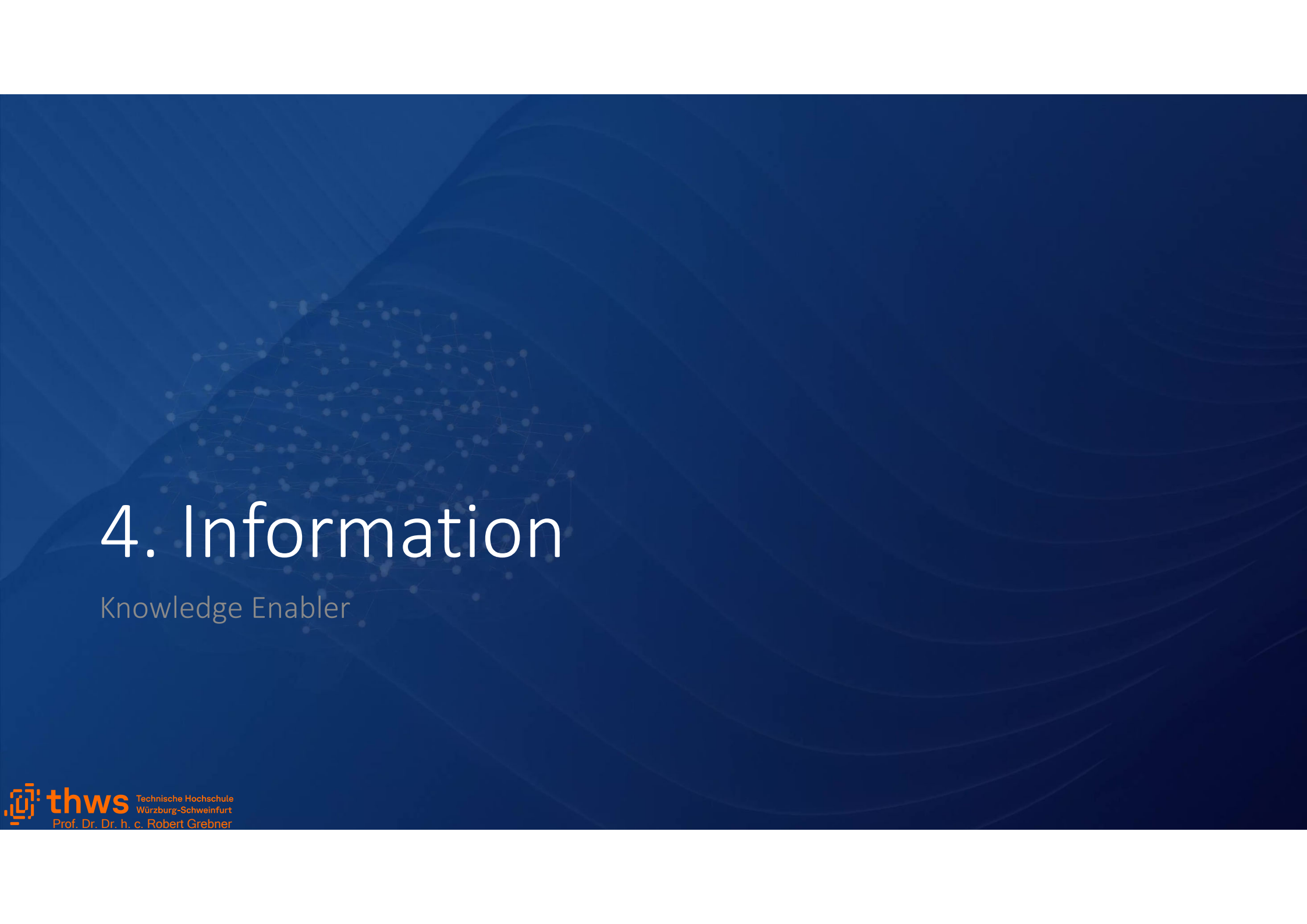


What is information needed for?

- to have a representation of the world we are living in
- needed to solve daily problems and survive
- ...

Communication is not information!

- Information is transferred/communicated to other thinkers on a physical base using media channels to make them able to get knowledge
- Information is materialised on media channels to be physically perceivable for other individuals
- by communication information leaves the mental world and enters it again (a materialised copy is leaving and new information is created)



4. Information

Knowledge Enabler

4. Information

Fundamentals

What is Information?

Mental Entity.

Is Information always
Mental?

Yes.

What is a Mental Entity?

A Representation.

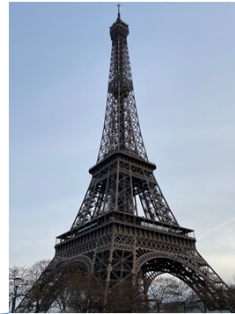
Where is Information?

Inside Brains.

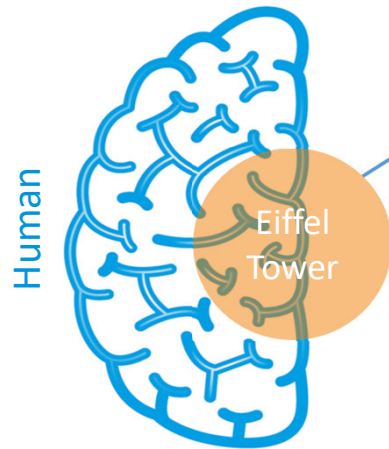
4. Information

Comparison Human and Machine Perception and Impression

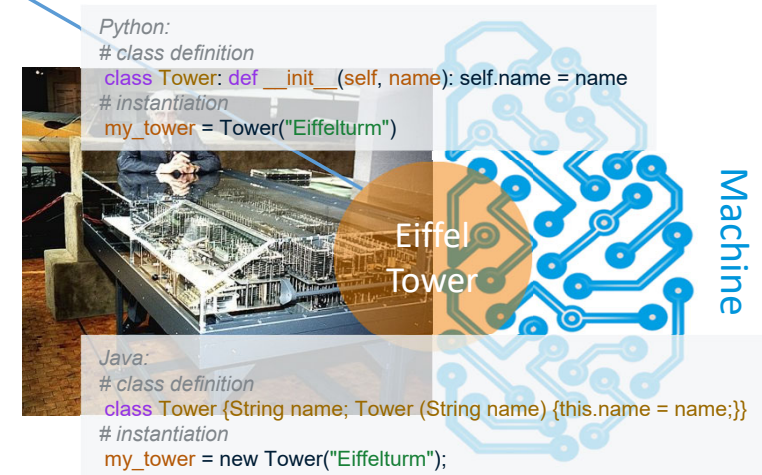
Physical World



Mental World



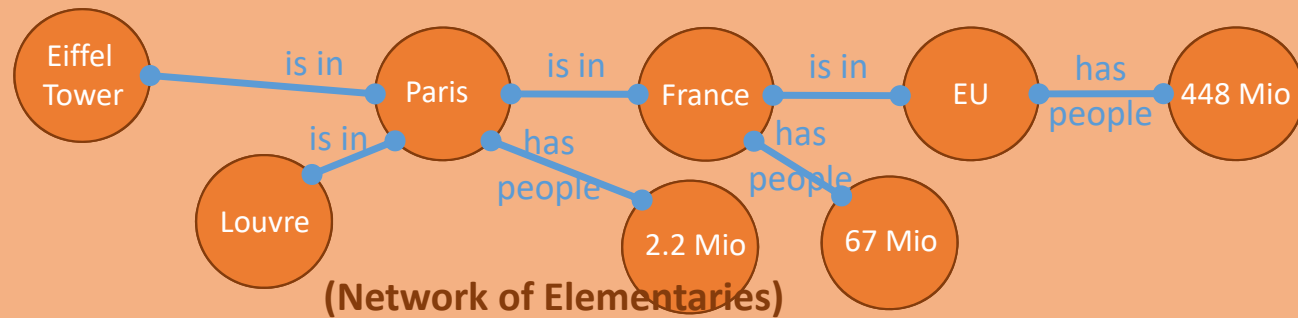
Mental Entity
=
Mental Representation
=
Information



4. Information

Granularity

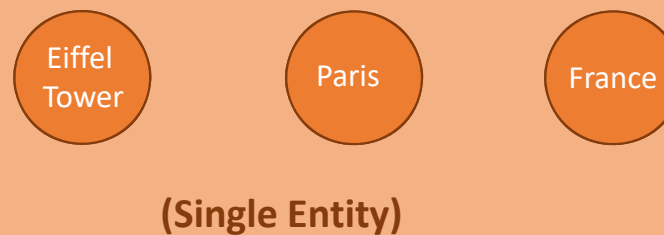
Complex Information



Elementary Information



Atomic Information



4. Information

Basic Information Characteristics

Entity Essence?

Understanding of an Entity
with its Complexity.

Entity Identifier?

Unique name.

Entity Type?

Category or class.

Entity Name?

Common Identifier.

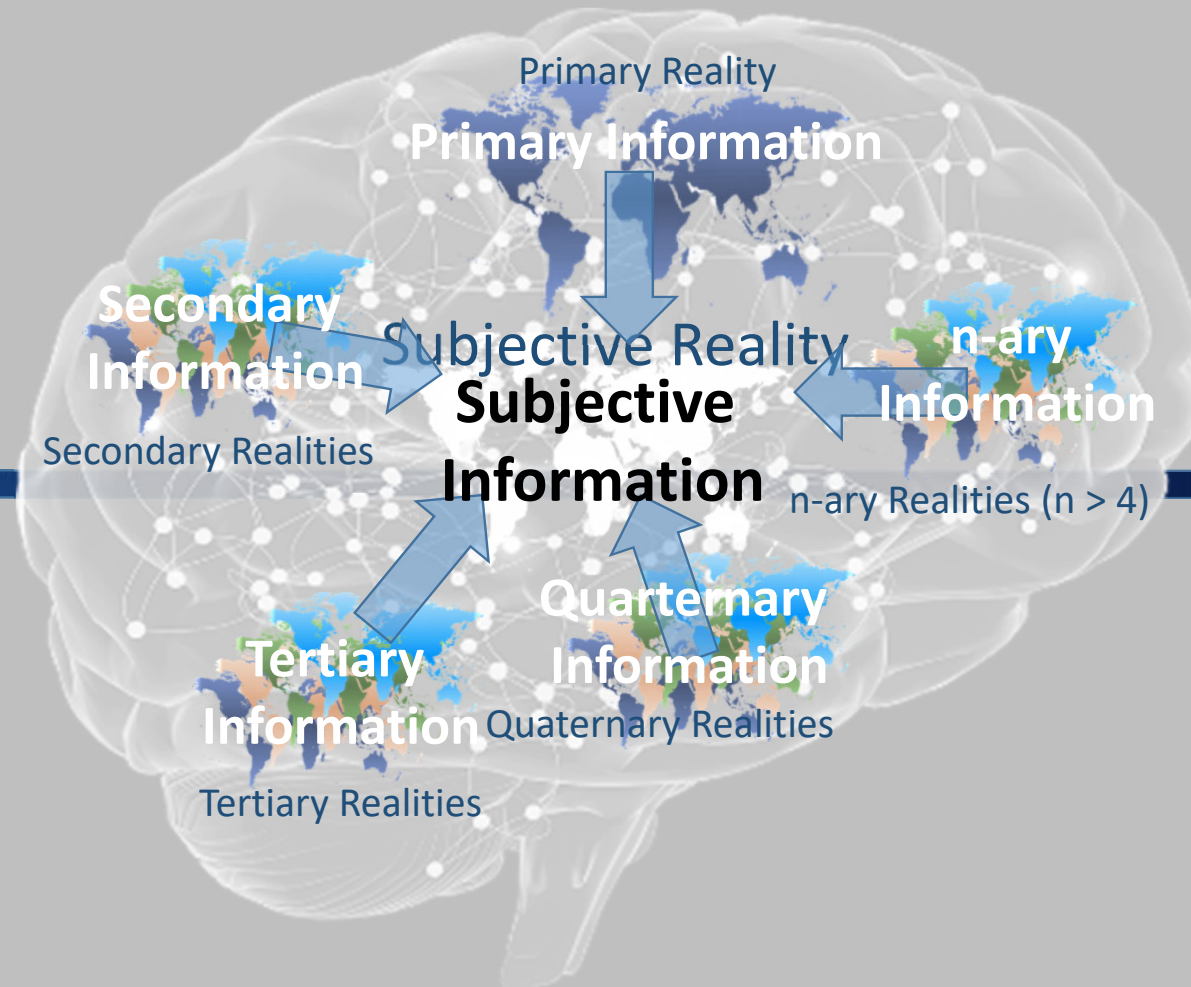
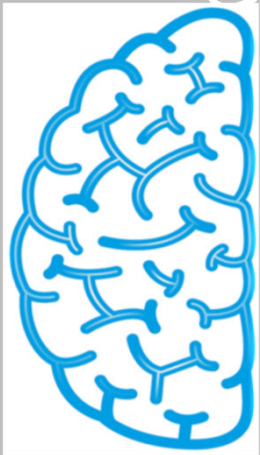
4. Information

Degree / Labelling

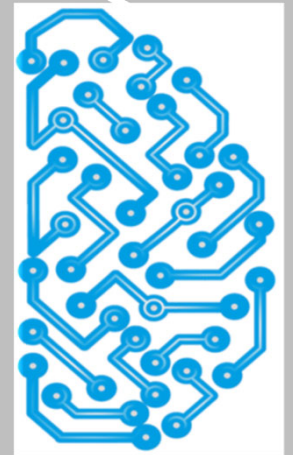
Physical World

Objective

Mental World



Reality





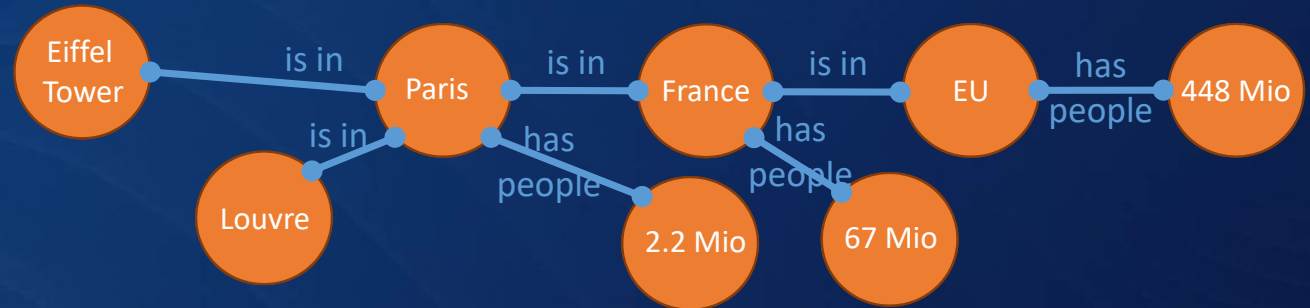
5. Knowledge

Action Enabler

Workshop „Knowledge“

- What was considered as being intelligent?
- Why is just information not sufficient to solve challenges?
- What makes an information valuable for solving challenges?
- Why do we learn? Why do we want to know more things?
- Where do we know from, what we want to learn?
- Where does the motivation comes from?

5. Knowledge



Knowing key figures of the European Union does not mean having knowledge, but information. As long, as you do not know what do do with that information, it is relatively useless for you. As soon as you can derive an purposeful acting (intelligence), you can say, it is knowledge. That means, mostly you have to connect that information network to other networks in a clever way.

DUDEN

Intelligenz, die

Substantiv, feminin

Fähigkeit [des Menschen], **abstrakt** und **vernünftig** zu **denken** und daraus **zweckvolles Handeln** abzuleiten

Ability [of human beings], to **think abstract** as well as **reasonable** and derive **purposeful acting** out of that

5. Knowledge

Fundamentals

What is the Base of Knowledge?

Information.

How what is Knowledge the Base?

Reasonable Thinking.

What is Knowledge needed for?

Purposeful Acting.

Is Thinking Acting?

Yes. Mental Acting.



6. Knowledge Gap

Motivation

Workshop „Knowledge Gap“

- How can we know about knowledge gaps (consider the information theory)?

6. Knowledge Gap

Concept Schemas

Through our whole live we learn a lot of concept schemas:

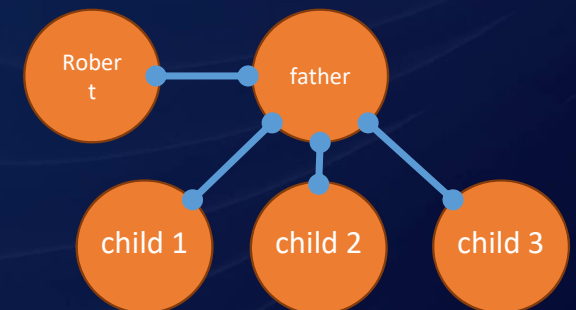
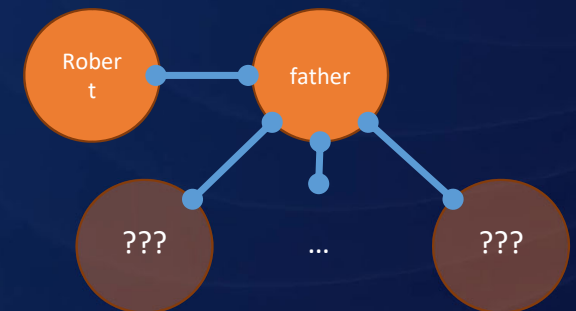
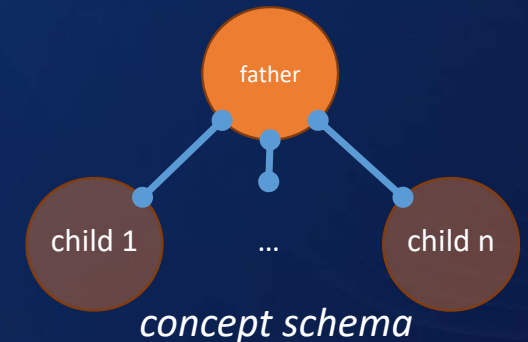
- Example concept „father“ with schema „father has one or more childs“ (that means one or more associations or elementary information units)

Once we hear, that a person Robert is a father, we immediately

- think of the imprinted/learned concept scheme „father“ and
- detect the information gap, which is the missing atomic child information
- we are used to ask for the number of childs and for the childs

We are used to ask for the number of childs and for the childs automatically to fill the knowledge gap. The answers could be:

- „He has three childs“



6. Knowledge Gap

Fundamentals

What is a Knowledge Gap?

The Information we Need to
Fullfill a Task.

Why do we Recognise Gaps?

We want to complete
uncomplete schemas.

When do we Recognise
Gaps?

While Making Action Plans.

When do we Look for
Concept Schemas?

When we Fail in Solving a
Problem.



7. Conclusion

Mental Subject

7. Conclusion

- Mental entities (information) are much more complex than physical entities.
- The complexity is about the representation of changes in the physical world including expressions of other individuals.
- Information uses „complexity reduction operators“ to handle all the happenings in the world.
- Information is inside thinking apparatus, but can also be materialised
- Information is always linked to a reality and in this sense is part of a specific reality.
- Knowledge is a set of information, that is needed for purposeful acting (Intelligence).
- Experiences (training on base of knowledge) help a lot to apply knowledge properly.



I
Thank
You